

numba-cuda vs numba-cuda-mlir

The question: does the choice of CUDA compiler backend change performance — and where? **The method:** profile the identical Stable-Fluids GPU solver (one source file, 9 kernels) built by both compilers, holding the GPU, CUDA 13.3 toolchain, workload, and Python version constant — so any difference is the compiler alone.

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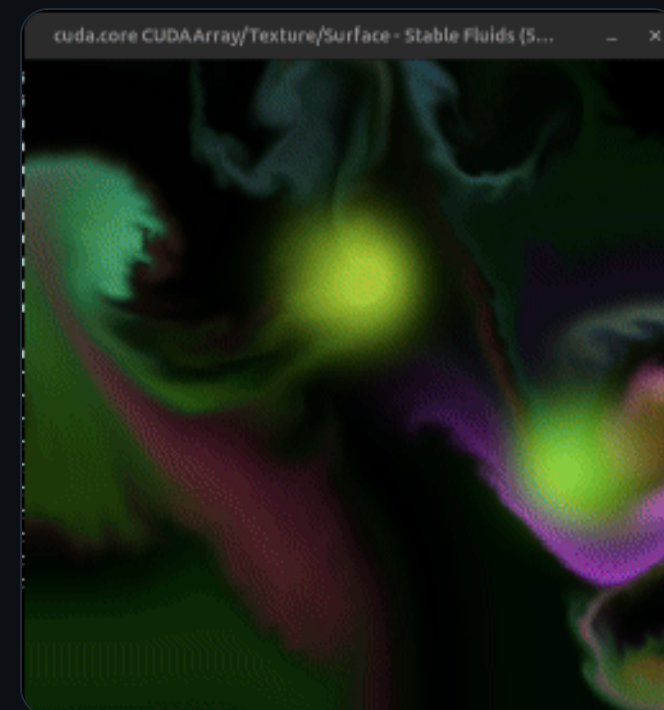
COMPILERS · 1 SOURCE

9KERNELS · 256^2 – 2048^2 **5**

MEASUREMENT AXES

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the workload – 512^2 Stable-Fluids

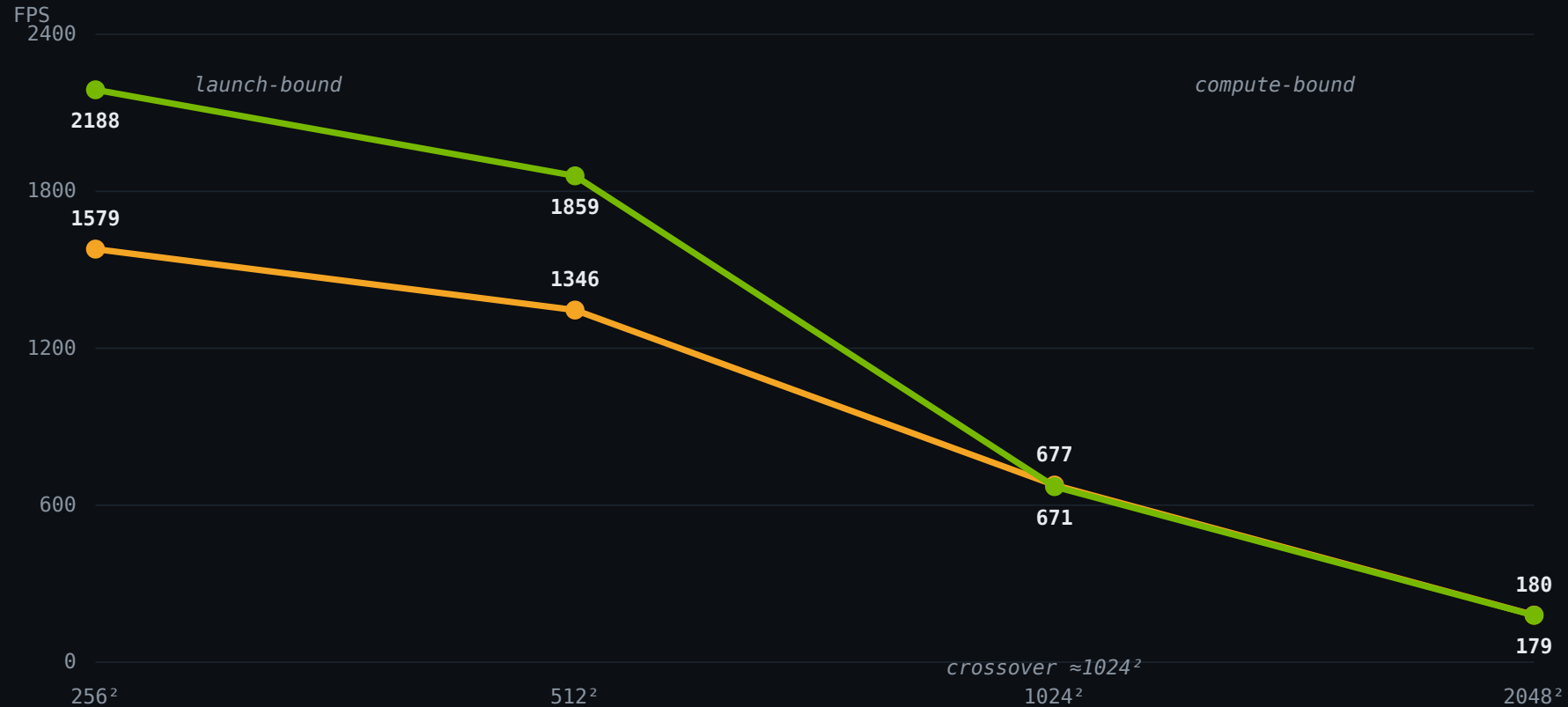
One variable: the compiler frontend

	CLASSIC	MLIR
package	numba-cuda 0.30.2	numba-cuda-mlir 0.4.0
frontend	Numba IR → LLVM → NVVM → PTX	MLIR → PTX (LTOIR)
opt / LTO default	NVVM opt-3, no LTO	LTO, opt>0
Python	3.12.13	3.12.13
cuda-bindings / ptxas	13.3.1 / CUDA 13.3	13.3.1 / CUDA 13.3
GPU	RTX 5880 Ada (sm_89), driver 595.71.05	

Bit-exact output on all 9 kernels at 256^2 / 512^2 / 1024^2 / 2048^2 → a fair race on identical work.

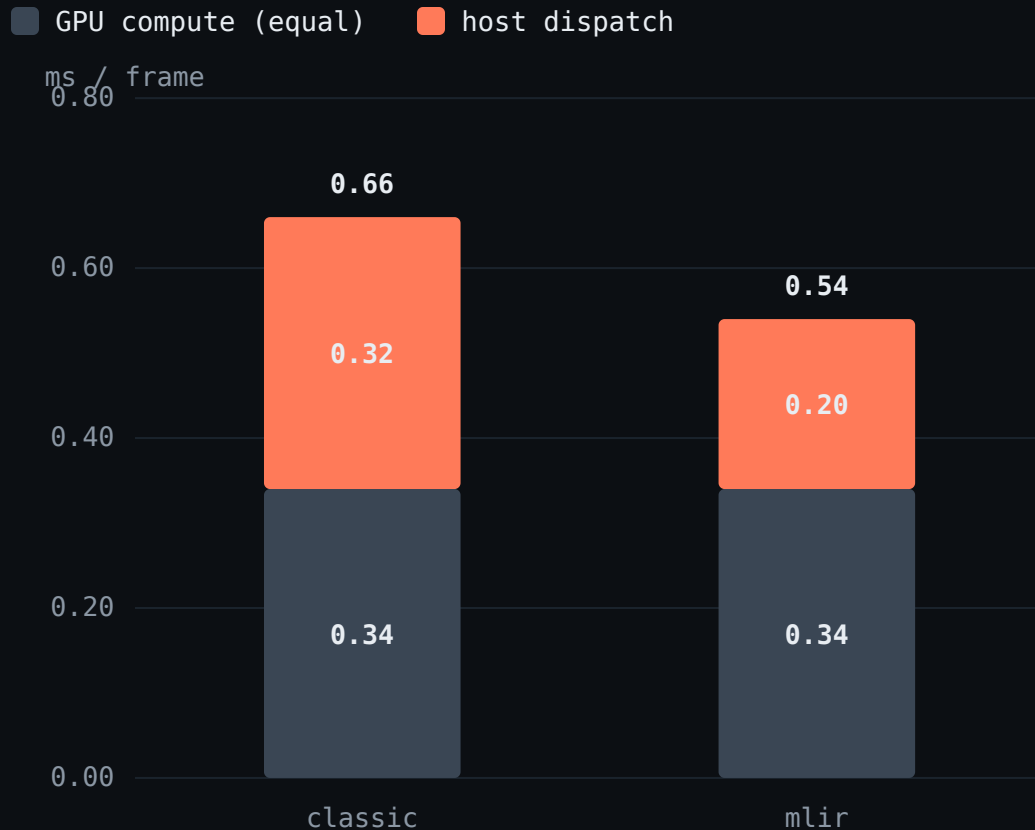
The launch-bound → compute-bound crossover

■ classic (NVVM) ■ mlir (MLIR)



Small grids = **host-launch-bound** (37 launches/frame): mlir wins **~38%**. Big grids = GPU-bound: **identical**.

Equal GPU time; unequal dispatch cost



- ▶ The steel base — GPU compute — is the **same 0.34 ms** in both.
- ▶ The coral top — host dispatch — differs: **8.6 μ s** vs **5.3 μ s** per launch.
- ▶ $\times 37$ launches/frame builds the total; mlir is **$\sim 38\%$ lighter** to dispatch.
- ▶ **Why:** classic runs $\sim 2\times$ the Python per launch (**69** vs **34** calls) — re-resolving the CUDA context and re-prepping args every time; mlir caches both.

Where classic spends the extra per-launch time

1. Re-resolve the CUDA context – every launch

```
# cudadriv/devices.py
with self._lock:
    with driver.get_active_context() as ac: # driver query, each call
        ...
```

2. Re-prepare every argument

```
# dispatcher.py _prepare_args
for ext in reversed(self.extensions): # hook loop
    ty, val = ext.prepare_args(...)
    devarg = wrap_arg(val).to_device(retr, stream) # re-wrap + convert
```

3. Safety-wrapped driver calls (2×) + cufunc lookup

```
# cudadriv/driver.py – twice per launch
def safe_cuda_api_call(*a):
    return self._check_cuda_python_error(fname, libfn(*a))
cufunc = self._codelibrary.get_cufunc() # lookup each launch
```

mlir – cached fast path

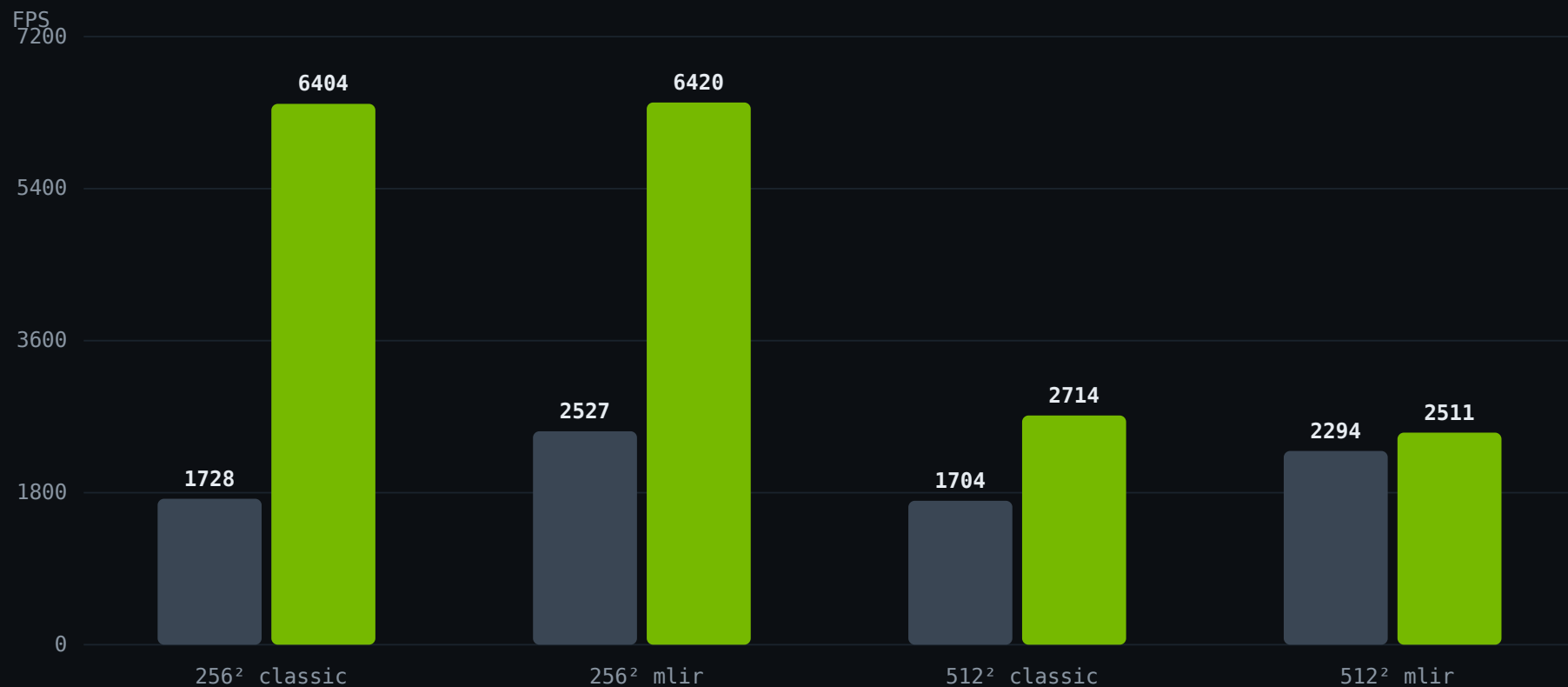
```
# numba_cuda_mlir/descriptor.py
type_key = tuple(type(a) for a in args)
cached = self._sig_cache.get(type_key)
if cached and fast_ok:
    return self._launch(argtypes, args) # skip re-prepare
```

69 vs 34

PYTHON CALLS / LAUNCH · CLASSIC VS MLIR

CUDA graphs erase the gap

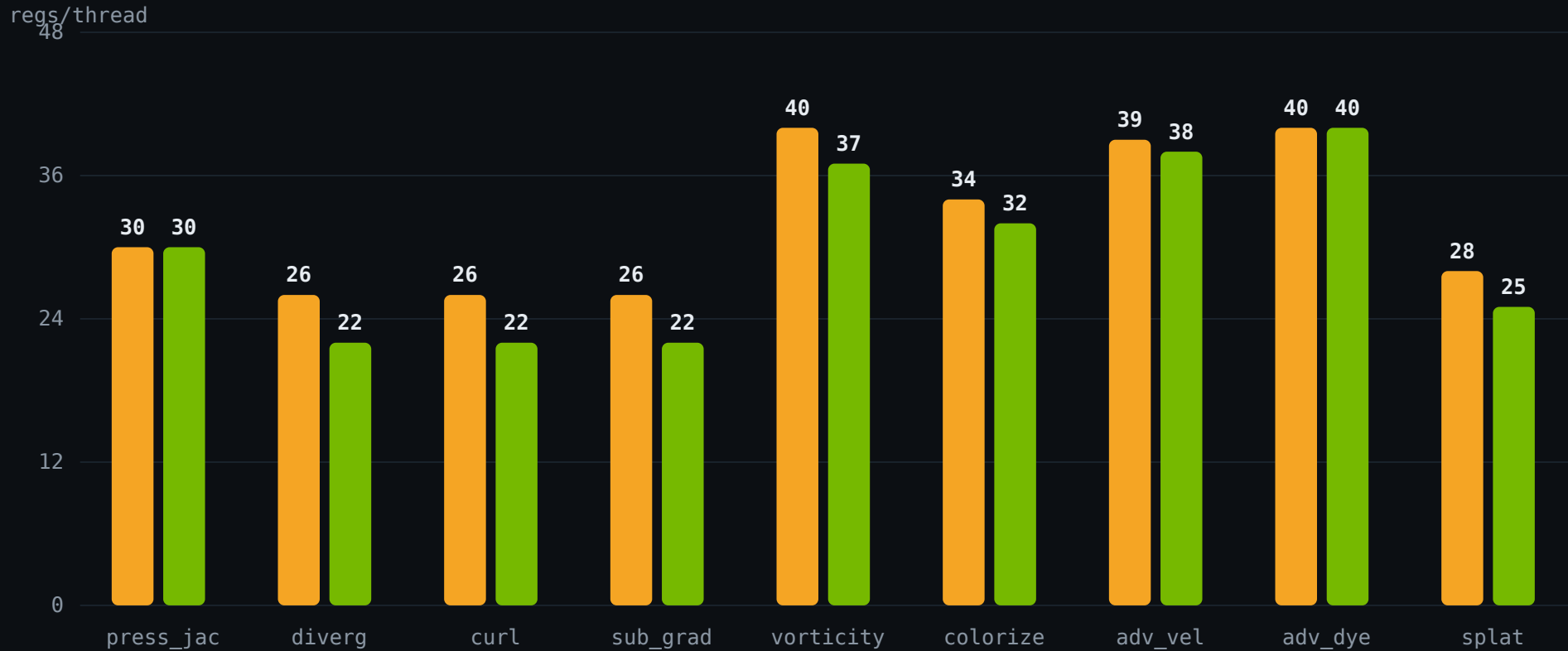
■ eager ■ graphed



Capture once, replay as a single launch. Both **converge to the same FPS** – classic gains more (**3.7×** vs 2.5×). mlir's eager edge **disappears**.

Slightly fewer registers — and it doesn't matter

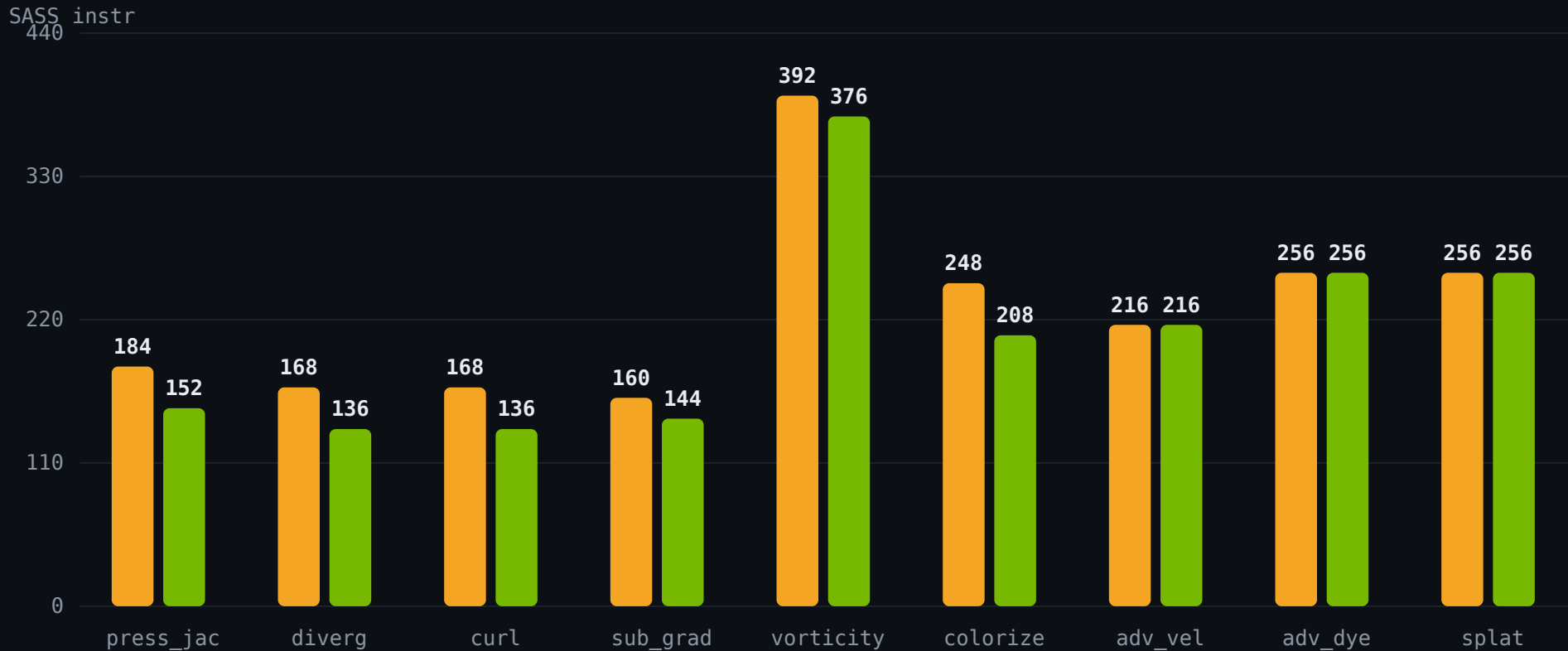
■ classic ■ mlir



All below the 42-reg occupancy limit → no occupancy change. Dominant `pressure_jacobi`: 30 = 30.

Fewer instructions $\neq$ faster here

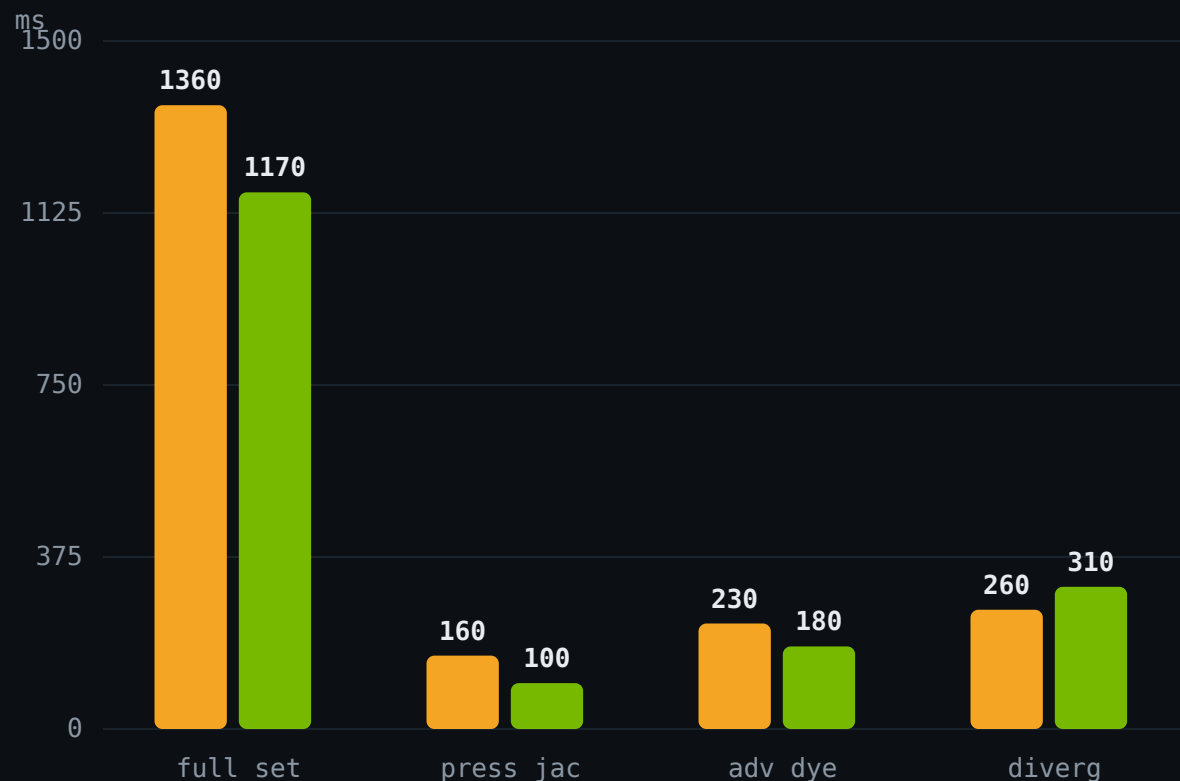
classic mlir



mlir trims -10-19% on stencils, classic wins colorize – yet GPU time is identical. The limiter is **float64 throughput** (Python literals promote), not instruction count.

mlir compiles ~14% faster

classic mlir



1.36s

CLASSIC – FULL SET

1.17s

MLIR – FULL SET (3.12)

Context pre-warmed → pure compile. No on-disk cache, so paid every process.

Match the settings, and the compilers tie

REGISTERS / THREAD pressure_jacobi · fewer = better	LTO OFF	LTO ON
classic	30	30
mlir	50	30

- ▶ **LTO** (link-time optimization) is a knob **both** compilers have.
- ▶ mlir turns it **on** by default; classic leaves it **off** — that's the whole reason mlir's code looked tighter on slides 6–7.
- ▶ Turn LTO on for both → **identical, 30 regs**. Neither compiler is fundamentally better.

A tie on everything — except how fast they launch kernels

WHAT WE MEASURED	WINNER	BY HOW MUCH
output correctness	tie	exact same numbers, every grid size
GPU kernel speed	tie	same runtime & occupancy
GPU code size	tie*	mlir writes ~15% fewer instrs – but no faster; equal once both use LTO
compile time	mlir	~14% faster
kernel-launch overhead	mlir	5.3 vs 8.6 μ s/launch $\rightarrow$ +38% FPS on small grids
FPS on big grids / with graphs	tie	identical

Core idea: for performance the two compilers are interchangeable – same results, same kernel speed. mlir's **only** real edge is cheaper kernel launching, which matters on small workloads and **vanishes with CUDA graphs**.

Pick on launch model, not the compiler

- ▶ **Same results, same kernel speed** — identical output; equal GPU time and occupancy.
- ▶ **mlir launches kernels ~38% cheaper** — 5.3 vs 8.6 μ s $\rightarrow$ +38% FPS below 1024².
- ▶ **CUDA graphs $\approx 3\times$ FPS** on both backends — and erase the mlir-vs-classic gap.
- ▶ **mlir compiles ~14% faster** — 1.17 s vs 1.36 s for all 9 kernels.
- ▶ **Biggest latent win:** kernels run in float64 (Python literals); float32 would cut math on both.